IoT BASED-INDOOR FIRE FIGHTER NAVIGATON AND SAFETY SUIT

Submitted in partial fulfilment of the requirements for the award of Bachelor of Engineering Degree in Electronics and Communication Engineering

by

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BONAFIDE CERTIFICATE

This is to certify that this project report is the Bonafide work of **P SHANMUKHA SAI MANIDEEP (43130298), P SATHISH (43130325), P DHANUSH (43130326), B SRI SAI JAYANTH (43130855)** who carried out the EMBEDDED SYSTEMS entitled "**IoT BASED INDOOR FIRE FIGHTHER NAVIGATION AND SAFTEY SUIT**" under my supervision from Jan 2026 to May 2026.

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ABSTRACT

Firefighters operate in extremely dangerous environments such as smoke-filled and low-visibility indoor spaces, where risks like toxic gas exposure, high temperatures, and physical exhaustion can threaten their lives. This project, titled "IOT-Based Indoor Firefighter Navigation and Safety Suit," is designed with a primary focus on enhancing firefighter safety while enabling effective indoor navigation and real-time monitoring. The system is built around the ESP32-WROOM-32, which integrates multiple sensors to continuously track both environmental conditions and the firefighter's health status. A DHT11 sensor monitors temperature and humidity, while the MQ-2 gas sensor detects hazardous gases and smoke levels. A heart rate sensor ensures real-time health monitoring, and the MPU6050 sensor detects motion, orientation, and possible falls, enabling immediate alerts in case of emergencies. A GPS module is used for location tracking, assisting rescue teams in identifying the firefighter's position. To further enhance safety, the system includes alert mechanisms such as a buzzer to warn the firefighter of dangerous conditions instantly. OLED display provides real-time data feedback.This safety suit provides continuous real-time data, helping to prevent accidents, detect life-threatening conditions early, and improve rescue response time. By combining navigation, health monitoring, and hazard detection, the system significantly enhances the overall safety, protection, and survivability of firefighters during indoor fire emergencies. This project, titled "IOT-Based Indoor Firefighter Navigation and Safety Suit," is designed with a primary focus on enhancing firefighter safety while enabling effective indoor navigation and real-time monitoring. The system is built around the ESP32-WROOM-32, which integrates multiple sensors to continuously track both environmental conditions and the firefighter's health status.

Keywords:

IoT-based, indoor firefighter navigation, safety suit, real-time monitoring, ESP32-WROOM-32, DHT11, MQ-2 gas sensor, heart rate sensor, MPU6050, fall detection, GPS module, hazardous gas detection, OLED display, buzzer alert, wireless communication, health monitoring, rescue response.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION OF THE PROJECT

Firefighting is one of the most hazardous professions, requiring individuals to operate in extreme and unpredictable environments. Firefighters are often exposed to high temperatures, toxic gases, dense smoke, and structurally unstable buildings. These conditions significantly reduce visibility and increase the risk of disorientation, injury, or even death. Traditional firefighting methods rely heavily on human experience and basic protective equipment, which may not be sufficient to ensure safety in all scenarios.

In indoor fire incidents, navigation becomes a major challenge due to thick smoke and limited visibility. Firefighters can easily lose their way, making rescue operations more difficult and time-consuming. Additionally, there is often no effective system to continuously monitor the firefighter's health conditions such as heart rate or physical stress levels during operations.

With the advancement of technology, especially in the field of the Internet of Things (IoT), it has become possible to design intelligent systems that can monitor environmental and physiological conditions in real time. Integrating sensors and communication technologies into a wearable system can greatly enhance the safety and efficiency of firefighters.

This project focuses on developing an IoT-based indoor firefighter navigation and safety suit that provides real-time monitoring, hazard detection, and communication capabilities to improve the overall safety of firefighters.

1.2 PROBLEM STATEMENT

Despite the availability of modern firefighting equipment, several challenges continue to affect the safety and performance of firefighters. One of the major issues is the lack of real-time monitoring of the firefighter's health status. During operations, firefighters are subjected to intense physical exertion and extreme environmental conditions.

which can lead to serious health risks that may go unnoticed without proper monitoring systems. Another significant problem is the difficulty in navigation within indoor environments filled with smoke and obstacles, where visibility is severely limited. This often results in firefighters losing their orientation, which can delay rescue efforts and increase the risk of accidents.

In addition to navigation challenges, firefighters are also exposed to harmful gases such as carbon monoxide and other toxic substances that are not easily detectable without specialized sensors. Prolonged exposure to such gases can have severe health consequences.

Furthermore, in the event of an emergency such as a fall or loss of consciousness, it becomes difficult for rescue teams to locate and assist the firefighter quickly due to the lack of an effective tracking system. Communication is another critical issue, as conventional communication methods may fail or become unreliable in harsh environments. These limitations highlight the need for an advanced system that can provide continuous monitoring, real-time alerts, and reliable communication to enhance firefighter safety and operational efficiency.

1.3 OBJECTIVES OF THE PROJECT

The main objective of this project is to design and develop an intelligent safety suit that enhances the safety of firefighters during indoor fire incidents. The system aims to provide real-time monitoring of environmental conditions such as temperature, humidity, and the presence of hazardous gases, ensuring that firefighters are aware of potential dangers in their surroundings. At the same time, the project focuses on monitoring the physiological condition of the firefighter by tracking parameters such as heart rate, which can help in identifying signs of stress or health issues during operations.

Another important objective of the project is to enable effective navigation by providing location tracking capabilities, which can assist both the firefighter and rescue teams in identifying positions within indoor environments. The system is also designed to detect motion and orientation using sensors, allowing it to identify incidents such as falls or lack of movement, which may indicate an emergency situation. Additionally, the project aims to implement alert mechanisms such as buzzer and vibration systems to provide immediate warnings to the firefighter when dangerous conditions are detected. Wireless communication is also incorporated to ensure that real-time data can be transmitted to remote monitoring stations, enabling quick decision-making and response. Overall, the project seeks to integrate multiple technologies into a single system that improves safety, awareness, and efficiency in firefighting operations.

1.4 MOTIVATION OF THE PROJECT

The motivation of this project is centered on the design and development of a prototype IoT-based safety suit for firefighters, with a focus on indoor environments where navigation and monitoring are particularly challenging.

The system integrates various sensors and modules into a wearable platform that can continuously collect and transmit data related to environmental conditions and the firefighter's health status. The project involves the use of a microcontroller-based system to process sensor data and generate appropriate alerts when necessary.

The developed system is intended to demonstrate the feasibility and effectiveness of using IoT technology to enhance firefighter safety. It includes features such as real-time monitoring, hazard detection, alert generation, and communication capabilities.

However, the scope of the project is limited to a prototype implementation and does not cover large-scale deployment or integration with advanced technologies such as detailed indoor mapping or artificial intelligence-based navigation systems. Despite these limitations, the project provides a strong foundation for further development and improvement in the field of smart firefighting solutions.

The shows an IoT-based Indoor Firefighter Navigation and Safety Suit designed to improve firefighter safety during emergency operations. The system uses a wearable smart suit equipped with multiple sensors such as MQ-2 gas sensor for detecting smoke and harmful gases, MPU6050 motion sensor for movement and fall detection, DHT11 for temperature and humidity monitoring, pulse oximeter sensor for heart rate and oxygen level, and a GPS module for live location tracking. All sensor data is collected and processed by the ESP32 microcontroller, which acts as the main control unit.

Based on the sensor readings, the system provides alerts using red and green LEDs, a buzzer, and an OLED display to show important information. It also sends real-time data through Wi-Fi and Bluetooth to a cloud server, mobile device, or control room for remote monitoring. In dangerous conditions such as fire, gas leakage, or health emergencies, alerts are immediately generated so rescue teams can respond quickly. This smart system helps ensure maximum safety, fast communication, and better navigation for firefighters indoors.

CHAPTER 2

LITERATURE SURVEY

2.1 MAJOR FINDINGS FROM LITERATURE SURVEY

The rapid advancement of technology in recent years has significantly influenced the development of intelligent safety systems, particularly in high-risk domains such as firefighting. The integration of the Internet of Things (IoT). wireless sensor networks, and embedded systems has opened new possibilities for enhancing the safety, efficiency, and operational effectiveness of firefighters. Researchers have extensively explored various approaches to improve fire detection, hazard monitoring, and emergency response systems.

This chapter presents a comprehensive review of existing literature related to IoT-based fire detection systems, wearable safety devices, firefighter monitoring technologies, and indoor navigation systems. The purpose of this review is to analyze existing solutions, identify their limitations, and establish the need for an advanced integrated system such as the proposed IoT-based indoor firefighter navigation and safety suit.

2.2 IoT IN FIRE SAFETY SYSTEMS

The application of IoT technology in fire safety systems has revolutionized traditional fire detection methods by enabling real-time monitoring and faster response mechanisms. Conventional fire detection systems often suffer from delayed response times and lack of communication capabilities.

However, IoT-based systems utilize interconnected sensors and communication networks to detect fire-related parameters such as temperature, smoke, and gas concentration in real time. According to recent studies, IoT-enabled fire detection systems significantly improve response time, reduce false alarms, and provide remote monitoring capabilities, making them more efficient and reliable compared to traditional systems.

In modern IoT-based fire detection systems, multiple sensors are integrated with microcontrollers to continuously monitor environmental conditions. When abnormal conditions are detected, the system triggers alerts and sends notifications to users through the internet. This real-time data transmission allows authorities to take immediate action, thereby reducing damage and saving lives. Recent research also highlights that IoT systems can be integrated with cloud platforms and mobile applications, enabling remote access and control of fire safety systems.

2.3 PROBLEM STATEMENT

Firefighting is one of the most critical and hazardous professions, where personnel are required to operate in highly unpredictable and life-threatening environments. During fire emergencies, especially in indoor settings such as residential buildings, industrial facilities, and commercial complexes, firefighters face numerous challenges that significantly impact their safety and efficiency. One of the major issues encountered during such operations is the presence of dense smoke, which drastically reduces visibility and makes navigation extremely difficult. In such conditions, firefighters can easily lose their sense of direction, leading to delays in rescue operations and increasing the risk of getting trapped inside hazardous zones. The absence of reliable indoor navigation systems further aggravates this problem, making it difficult for both firefighters and rescue teams to track positions accurately.

Another critical concern is the exposure to extreme environmental conditions, including high temperatures, humidity, and the presence of toxic gases such as carbon monoxide and other harmful substances. These gases are often invisible and odorless, making them difficult to detect without specialized equipment. Prolonged exposure to such hazardous conditions can lead to severe health issues, including respiratory problems, unconsciousness, or even death. Traditional firefighting suits primarily provide passive protection against heat and flames but lack the capability to actively monitor environmental parameters in real time. This limitation prevents firefighters from being aware of dangerous conditions before they reach critical levels.

In addition to environmental risks, firefighters are also subjected to intense physical stress during operations. Carrying heavy equipment, working in high-temperature environments, and performing physically demanding tasks can lead dehydration, and cardiovascular strain. However, there is often no system in place to continuously monitor the physiological condition of firefighters, such as heart rate or physical activity levels. As a result, early signs of health issues may go unnoticed, increasing the likelihood of sudden medical emergencies during operations. This lack of real-time health monitoring poses a significant threat to firefighter safety.

Furthermore, emergency situations such as falls, injuries, or loss of consciousness are common in firefighting scenarios due to unstable structures, slippery surfaces, and limited visibility. In such cases, immediate assistance is crucial to prevent serious consequences. However, without an effective system to detect these incidents and alert rescue teams, response time can be significantly delayed. The absence of motion detection and automatic alert mechanisms makes it difficult to identify and respond to such emergencies promptly.

Communication is another major challenge in firefighting operations. Conventional communication systems may become unreliable or fail completely in extreme conditions due to interference, structural obstacles, or damage to infrastructure. This can lead to a breakdown in coordination between team members and control units, making it difficult to share critical information in real time. Effective communication is essential for ensuring coordinated efforts, quick decision-making, and timely response during emergencies, and its absence can severely impact the success of firefighting operations.

Moreover, locating firefighters inside buildings during emergencies remains a significant challenge. While global positioning systems (GPS) are widely used for outdoor navigation, their effectiveness is limited in indoor environments due to signal attenuation and interference. This makes it difficult for rescue teams to accurately determine the location of firefighters who may be trapped or in distress. The lack of a reliable tracking system further increases the risk associated with indoor firefighting operations.

Despite advancements in technology, many existing solutions address only specific aspects of these challenges, such as fire detection or gas monitoring, without providing a comprehensive approach to firefighter safety. The lack of integration between different systems limits their effectiveness in real-world scenarios, where multiple factors need to be monitored simultaneously.

2.4 PROBLEMS IN EXISTING METHOD

Despite the considerable progress made in the development of fire safety technologies, several open problems continue to limit the effectiveness of existing systems, particularly in the context of indoor firefighting operations. One of the primary issues is the lack of integration among various subsystems. Many currently available solutions are designed to address specific aspects such as fire detection, gas monitoring, or communication, but they operate independently rather than as part of a unified system. This fragmented approach results in limited situational awareness, as firefighters and control units are unable to access comprehensive real-time information from a single platform. The absence of an integrated system that combines environmental monitoring, physiological tracking, navigation, and communication remains a significant gap in existing technologies.

Another major challenge lies in the reliability and accuracy of sensors under extreme conditions. Fire environments are characterized by high temperatures, dense smoke, and the presence of corrosive gases, all of which can degrade sensor performance. Many conventional sensors are not designed to withstand such harsh conditions for extended periods, leading to inaccurate readings or complete sensor failure. This unreliability can result in false alarms or missed detections, both of which can have serious consequences during emergency operations. Ensuring consistent and accurate sensor performance in real-world firefighting scenarios continues to be an unresolved issue.

Indoor navigation also remains a critical open problem. While technologies such as GPS are effective for outdoor tracking, they perform poorly in indoor environments due to signal attenuation and obstruction. Existing indoor positioning systems often rely on additional infrastructure such as beacons or pre-installed networks, which may not be available or functional during fire incidents. As a result, accurately determining the location of firefighters inside buildings is still a challenging task. The lack of a robust, infrastructure-independent indoor navigation system significantly affects rescue operations and increases the risk of firefighters becoming disoriented or trapped.

Many current systems depend on standard communication technologies that are not optimized for such extreme conditions, leading to intermittent or complete loss of connectivity. This lack of reliable communication hinders coordination among team members and delays the transmission of critical information, thereby reducing the overall effectiveness of emergency response.

Power management is also a significant concern in wearable safety systems. Most existing devices rely on battery power, but continuous operation of multiple sensors and communication modules can quickly drain the battery. In firefighting scenarios, where operations can last for extended periods, maintaining a reliable power supply is crucial. However, current solutions often struggle to balance performance and energy efficiency, leading to limited operational time and reduced reliability.

Another unresolved issue is the lack of advanced data processing and intelligent decision-making capabilities in many systems. While some systems are capable of collecting and transmitting data, they do not effectively analyze this data to provide meaningful insights or predictive warnings. The absence of intelligent algorithms for real-time analysis limits the ability of these systems to anticipate dangerous situations and take proactive measures. This highlights the need for smarter systems that can not only monitor conditions but also interpret data and assist in decision-making.

Furthermore, user comfort and ergonomics are often overlooked in the design of wearable systems. Firefighters already carry heavy protective gear, and the addition of bulky or poorly designed electronic systems can hinder their movement and performance. Existing solutions may not adequately consider factors such as weight, flexibility, and ease of use, which are essential for practical deployment in real-world scenarios.

2.5 DISADVANTAGES OF THE EXISTING SYSTEM

Existing firefighting safety and monitoring systems, although improved over time, still suffer from several significant disadvantages that limit their effectiveness in real-world emergency situations. One of the primary drawbacks is the lack of a unified and integrated system. Most currently available solutions are designed to perform specific functions such as fire detection, gas monitoring, or communication, but they operate independently rather than as a coordinated system. This fragmented approach makes it difficult to obtain a complete and real-time understanding of the situation, which is critical during firefighting operations.

Another major disadvantage is the limited capability of traditional firefighting suits and equipment. Conventional safety gear mainly provides passive protection against heat and flames but does not include active monitoring features. This means that firefighters are not continuously informed about environmental hazards such as rising temperature levels or the presence of toxic gases. Without real-time feedback, they may unknowingly enter dangerous zones, which can lead to severe health risks or fatal incidents. The absence of intelligent alert mechanisms further increases the likelihood of delayed responses to critical situations.

Sensor reliability is also a major concern in existing systems. Many sensors used in fire safety applications are not specifically designed to withstand extreme conditions such as high temperatures, heavy smoke, and exposure to corrosive gases. Under such harsh environments, sensors may produce inaccurate readings or fail completely. This can lead to false alarms or missed detections, both of which are dangerous in emergency scenarios. Inaccurate data can mislead firefighters and decision-makers, ultimately compromising safety and operational effectiveness.

Indoor navigation remains another significant limitation. Most existing systems rely on external positioning technologies such as GPS, which are ineffective in indoor environments due to signal loss and interference. As a result, firefighters often lack reliable location tracking when operating inside buildings. This makes it difficult for them to navigate through complex structures and increases the risk of getting lost. Additionally, rescue teams face challenges in locating firefighters who may be trapped or in distress, leading to delays in providing assistance.

Communication issues further add to the disadvantages of existing systems. In many firefighting scenarios, communication networks become unreliable due to structural damage, interference, or environmental conditions. Traditional communication devices may fail to provide consistent connectivity, making it difficult for team members to coordinate their actions effectively. The lack of a robust communication system can lead to miscommunication, delayed decision-making, and reduced operational efficiency during critical situations.

Another important drawback is the absence of real-time health monitoring. Firefighters are exposed to intense physical strain, high temperatures, and stressful systems do not include mechanisms to monitor physiological parameters such as heart rate or body condition. This lack of monitoring makes it difficult to detect early signs of fatigue, stress, or medical emergencies, increasing the risk of sudden health-related incidents during operations.

Power consumption and battery limitations also pose significant challenges. Many existing electronic systems require continuous operation of sensors and communication modules, which consume a considerable amount of power. In firefighting situations, where operations can last for extended periods, maintaining sufficient battery life becomes critical. However, current systems often struggle with efficient power management, leading to reduced operational time and the possibility of system failure when it is needed most.

Furthermore, existing systems often lack intelligent data processing capabilities. While some systems are capable of collecting data, they do not effectively analyze it to provide meaningful insights or predictive warnings. This limits their ability to support decision-making and prevents proactive measures from being taken in dangerous situations. Without advanced processing and analysis, the full potential of collected data remains underutilized.

The design and ergonomics of existing wearable systems also present disadvantages. Many devices are bulky, heavy, or difficult to use, which can hinder the movement and performance of firefighters. Since firefighters already carry heavy protective gear, adding additional equipment without proper design considerations can increase physical strain and reduce efficiency. Ease of use and comfort are critical factors that are often overlooked in existing solutions.

Finally, the cost and complexity of advanced firefighting systems can limit their widespread adoption. Many high-end solutions are expensive and require specialized infrastructure, making them inaccessible for smaller organizations or regions with limited resources.

2.6 RESEARCH GAP IDENTIFICATION

The review of existing literature and currently available systems clearly indicates that significant advancements have been made in the fields of fire detection, IOT-based monitoring, wearable technology, and emergency response systems.

However, despite these developments, several critical gaps still exist that limit the effectiveness of these technologies in ensuring firefighter safety, particularly in indoor environments. These gaps highlight the need for a more comprehensive and integrated solution, which forms the basis for the proposed IOT-based indoor firefighter navigation and safety suit.

One of the most prominent research gaps is the lack of a fully integrated system that combines multiple functionalities into a single wearable platform. Most existing systems are designed to address specific aspects such as gas detection, temperature monitoring, or communication, but they operate independently without proper coordination. This separation results in fragmented data and limited situational awareness, making it difficult for firefighters and control units to make informed decisions in real time. There is a clear need for a unified system that can simultaneously monitor environmental conditions, track physiological parameters, detect motion, and provide navigation support within a single framework.

Another significant gap lies in the limited focus on real-time health monitoring of firefighters. While some research has explored wearable devices for monitoring physiological parameters, these solutions are often not integrated with environmental monitoring systems. Firefighters operate under extreme physical stress, and the absence of continuous health monitoring increases the risk of unnoticed medical emergencies. Existing systems do not adequately address the combined impact of environmental hazards and physical strain on the firefighter's body. Therefore, there is a need for a system that integrates both environmental and physiological monitoring to provide a comprehensive safety solution.

Indoor navigation remains an unresolved challenge and represents a major research gap in current systems. Most existing technologies rely on GPS for location tracking, which is ineffective in indoor environments due to signal attenuation and interference. Although some alternative methods such as beacon-based systems and inertial navigation have been proposed, they often require additional infrastructure or suffer from accuracy issues. There is a lack of reliable, cost-effective, and infrastructure-independent indoor navigation systems that can function effectively in fire-affected environments. Addressing this gap is essential for improving situational awareness and enabling efficient rescue operations.

The reliability and robustness of sensors in harsh firefighting conditions also present a critical gap. Many sensors used in existing systems are not specifically designed to withstand extreme temperatures, dense smoke, and exposure to toxic gases. As a result, their performance may degrade or fail during actual firefighting operations, leading to inaccurate data and unreliable system behavior. Current research does not sufficiently address the development of durable and high-performance sensors capable of operating consistently in such environments. This highlights the need for more robust sensor integration and validation under real-world conditions.

Another important gap is related to communication reliability. Although various wireless communication technologies have been implemented in existing systems, their performance in fire environments remains inconsistent. Factors such as interference, structural barriers, and network congestion can disrupt communication, leading to delays or loss of critical information. Many existing solutions do not provide mechanisms for ensuring reliable and continuous communication under such conditions. There is a need for more resilient communication strategies that can maintain connectivity and ensure seamless data transmission during emergencies.

Power management is also an area where current research is limited. Wearable safety systems require continuous operation of multiple sensors and communication modules, which results in high power consumption. Existing solutions often struggle to provide efficient energy management, leading to reduced battery life and limited operational duration. This is particularly problematic in firefighting scenarios, where prolonged operation is essential. There is a need for research focused on optimizing power consumption and developing energy-efficient systems that can operate reliably for extended periods.

2.7 SMART SENSORS FOR FIRE HAZARD DETECTION

Smart sensors play a crucial role in detecting fire hazards and ensuring safety in hazardous environments. Various types of sensors, such as temperature sensors, gas sensors, and smoke detectors, are widely used in fire detection systems. These sensors exceeded. Research indicates that combining multiple sensors improves the accuracy and reliability of fire detection systems, as it reduces false alarms and enhances detection efficiency.

A systematic review of IoT-based fire hazard systems highlights the importance of multi-sensor integration in smart buildings, where sensors are used for early fire detection, evacuation planning, and indoor navigation. The study identifies key themes such as vision-based sensing, automation, and intelligent decision-making, which contribute to improved fire safety systems.

2.7.1 WEARABLE TECHNOLOGY FOR FIREIFIGHTER SAFETY

Wearable technology has emerged as a promising solution for enhancing firefighter safety. Modern wearable devices are equipped with sensors that can monitor physiological parameters such as heart rate, body temperature, and movement. These devices provide real-time data, which can be used to assess the health and condition of firefighters during operations.

Recent studies have focused on developing wearable systems that integrate multiple sensors for monitoring both environmental and physiological conditions. For example, a multi-sensor IoT-based system has been developed to recognize firefighting activities using data from heart rate sensors, inertial measurement units, and other wearable sensors. The system demonstrated high accuracy in activity recognition, highlighting the potential of wearable technology in improving firefighter safety and performance .

Wearable safety systems are particularly useful in detecting emergency situations such as falls or lack of movement. By analyzing motion data, these systems can identify abnormal patterns and trigger alerts, enabling quick response from rescue teams. This capability is essential in hazardous environments where immediate assistance is required.

2.8 HAZARDOUS GAS MONITORING SYSTEMS

Fire environments often contain hazardous gases such as carbon monoxide, hydrogen cyanide, and other toxic substances that pose serious health risks to firefighters. Monitoring these gases is critical for ensuring safety and preventing life-threatening situations. Traditional firefighting suits provide passive protection, but recent advancements focus on active monitoring systems that can detect hazardous gases in real time.

A recent study on integrated firefighting suits demonstrates the use of advanced sensor systems combined with predictive models to monitor gas levels and provide early warnings. The system is capable of predicting hazardous gas trends and generating alerts before the situation becomes critical, significantly improving firefighter survival rates . This shift from passive protection to active monitoring represents a major advancement in firefighter safety technology.

2.8.1 WIRELESS COMMUNICATION AND IOT NETWORKS

Wireless communication is a fundamental component of IoT-based systems, enabling data transmission between sensors, processing units, and remote monitoring stations. Technologies such as Wi-Fi, Bluetooth, and wireless sensor networks are commonly used in fire safety systems to ensure reliable communication.

Research on wireless sensor networks highlights their importance in real-time monitoring and control of environmental parameters. These networks allow multiple sensors to communicate with a central system, providing continuous data updates and enabling remote monitoring. The use of wireless communication reduces the need for physical presence, improves operational efficiency, and enhances safety in hazardous environments .

However, communication reliability remains a challenge in firefighting environments due to interference, obstacles, and extreme conditions. Therefore, designing robust communication systems is essential for ensuring the effectiveness of IoT-based safety solutions.

2.8.2 INDOOR NAVIGATION SYSTEMS FOR FIREFIGHTERS

Indoor navigation is a critical aspect of firefighting operations, as firefighters often work in environments with limited visibility and complex layouts. Traditional navigation methods rely on experience and physical markers, which may not be sufficient in emergency situations.

Recent research has explored the use of IoT and sensor-based systems for indoor navigation. These systems utilize technologies such as GPS, inertial sensors, and wireless communication to track the location of firefighters and provide navigation assistance. Although GPS is less effective indoors, it can still be used in combination with other sensors to improve location accuracy.

Studies indicate that integrating navigation systems with fire detection and monitoring systems can significantly enhance situational awareness and improve rescue operations. By providing real-time location data, these systems enable faster response and reduce the risk of firefighters getting lost.

2.9 ROBOTICS IN FIREFIGHTING

Robotics has also been explored as a solution for reducing the risks associated with firefighting. IoT-based firefighting robots are designed to detect and extinguish fires without direct human involvement. These robots can operate in hazardous environments, reducing the exposure of firefighters to dangerous conditions.

Research on IoT-based firefighting robots highlights their ability to detect fire, identify its type, and apply appropriate extinguishing methods. These systems are often controlled remotely and can be integrated with IoT platforms for real-time monitoring and control . Although robotics provides significant advantages, it cannot completely replace human firefighters, making wearable safety systems equally important.

2.9.1 SMART CITY AND INTEGRATED FIRE SAFETY SYSTEMS

The concept of smart cities has further accelerated the development of advanced fire safety systems. In smart city environments, IoT-based systems are used to monitor and manage various aspects of urban infrastructure, including fire safety. These systems integrate multiple technologies such as sensors, communication networks, and data analytics to provide comprehensive safety solutions.

Research on integrated fire detection systems demonstrates the use of IoT and image processing techniques for accurate and real-time fire detection. These urban infrastructure enhances overall safety and supports efficient emergency response.

2.10 CHALLENGES IN EXISTING SYSTEMS

Despite significant advancements, existing fire safety systems still face several challenges. One of the major issues is the lack of integration between different components, such as sensors, communication systems, and monitoring platforms. Many systems operate independently, limiting their effectiveness in complex scenarios.

Another challenge is the reliability of sensors and communication networks in harsh environments. Factors such as high temperature, smoke, and interference can affect the performance of sensors and communication systems. Additionally, power consumption and battery life are critical concerns in wearable systems, as continuous operation is required during firefighting missions.

Security is also a concern in IoT-based systems, as vulnerabilities can lead to unauthorized access or system failures. Ensuring the security and reliability of IoT systems is essential for their successful implementation in critical applications.

2.11 NEED FOR PROPOSED SYSTEM

The literature review clearly indicates that while significant progress has been made in the development of IoT-based fire safety systems, there is still a need for a comprehensive solution that integrates multiple functionalities into a single system. Most existing systems focus on specific aspects such as fire detection, gas monitoring, or activity recognition, but lack a unified approach.

The proposed IoT-based indoor firefighter navigation and safety suit addresses these limitations by combining environmental monitoring, health tracking, navigation, and communication into a single wearable system. This integrated approach enhances situational awareness, improves safety, and provides a more effective solution for firefighting operations.

2.12 SUMMARY

In summary, the literature review highlights the importance of IoT, sensor technologies, wearable systems, and communication networks in improving fire safety and firefighter protection. Existing research demonstrates the potential of these technologies in enhancing detection, monitoring, and response capabilities. However, challenges such as system integration, reliability, and communication limitations still need to be addressed.

The proposed system builds upon the strengths of existing technologies while addressing their limitations, providing a comprehensive solution for improving firefighter safety and operational efficiency.

CHAPTER 3

SCOPE OF PRESENT INVESTIGATION

3.1 INTRODUCTON

The increasing complexity and risk associated with firefighting operations, particularly in indoor environments, demand the development of advanced technological solutions that can enhance safety, efficiency, and real-time decision-making. Traditional firefighting equipment primarily focuses on protection against heat and flames, but it lacks the capability to provide continuous monitoring of environmental and physiological conditions. In recent years, the emergence of the Internet of Things (IoT) and embedded systems has created new opportunities to design intelligent and connected safety solutions. These advancements enable the integration of sensors, communication modules, and processing units into wearable systems that can monitor, analyze, and respond to hazardous situations in real time.

The present investigation focuses on the design and development of an IoT-based indoor firefighter navigation and safety suit that aims to address the critical challenges faced by firefighters during emergency operations. The scope of this investigation lies in exploring how multiple sensing technologies and communication systems can be combined into a single, compact, and efficient platform to enhance firefighter safety. The system is intended to continuously monitor environmental parameters such as temperature, humidity, and the presence of hazardous gases, while also tracking the physiological condition of the firefighter through heart rate and oxygen level monitoring. By integrating these features, the system provides a comprehensive understanding of both external and internal risk factors.

In addition to monitoring, the investigation emphasizes the importance of real-time alerts and communication. The system is designed to detect abnormal conditions such as high gas concentration, extreme temperature, or lack of movement, and immediately notify the firefighter through alert mechanisms such as a buzzer and visual indicators. At the same time, the collected data is transmitted wirelessly to remote monitoring systems, enabling control rooms or rescue teams to track the firefighter’s condition and location. This real-time data exchange plays a crucial role in improving situational awareness and enabling faster and more effective response during emergencies.

Another important aspect of this investigation is the implementation of indoor navigation support. Since traditional positioning systems such as GPS have limitations in indoor environments, the study explores the use of available modules and sensor data to provide approximate location tracking and assist in rescue operations. The inclusion of motion sensing capabilities also allows the system to detect falls or inactivity, which may indicate that the firefighter is in distress. These features contribute to enhancing the overall safety and survivability of firefighters in hazardous conditions.

The scope of the present investigation is primarily focused on the development of a functional prototype using commonly available components such as gas sensors, motion sensors, humidity and temperature sensors, pulse oximeter sensors, GPS modules, and a microcontroller-based processing unit. The study also considers factors such as power management, system integration, and user interface design to ensure that the system is practical and efficient for real-world applications. However, the investigation is limited to prototype-level implementation and does not include large-scale deployment or advanced technologies such as artificial intelligence-based navigation or fully automated rescue systems.

Overall, this investigation aims to demonstrate the feasibility and effectiveness of integrating IoT technology into firefighter safety systems. By addressing key challenges such as hazard detection, health monitoring, communication, and navigation, the proposed system seeks to provide a reliable and comprehensive solution that enhances firefighter safety and operational efficiency in indoor fire emergencies.

3.2 OBJECTIVE

The primary objective of the present investigation is to design and develop an intelligent and integrated IoT-based indoor firefighter navigation and safety suit that enhances the safety, awareness, and operational efficiency of firefighters working in hazardous environments. The system aims to overcome the limitations of conventional firefighting equipment by incorporating real-time monitoring, alert mechanisms, and communication capabilities into a single wearable platform. This investigation focuses on creating a solution that not only protects firefighters from external hazards but also continuously monitors their internal health condition and provides timely alerts during critical situations.

A key objective of this study is to implement continuous environmental monitoring by integrating sensors capable of detecting parameters such as temperature, humidity, and the presence of hazardous gases. By doing so, the system ensures that firefighters are constantly aware of their surroundings and can identify dangerous conditions at an early stage. This helps in preventing exposure to life-threatening environments and allows for proactive decision-making during operations. Along with environmental monitoring, the investigation also aims to incorporate physiological monitoring by using a pulse oximeter sensor to track vital health parameters such as heart rate and blood oxygen levels. This feature is essential for identifying signs of fatigue, stress, or medical emergencies, thereby improving the overall safety of the firefighter.

Another important objective of the present investigation is to develop a reliable motion detection mechanism using appropriate sensors to identify the movement, orientation, and activity of the firefighter. This enables the system to detect abnormal situations such as falls or prolonged inactivity, which may indicate that the firefighter is in distress. In such cases, the system is designed to automatically generate alerts and notify both the firefighter and the control unit, ensuring a quick response. Additionally, the investigation aims to provide location tracking capabilities through the integration of a GPS module, which assists in identifying the approximate position of the firefighter and supports rescue operations.

The study also focuses on implementing an efficient alert system that can immediately notify the firefighter of dangerous conditions using visual and audio indicators such as LEDs and a buzzer. This ensures that critical information is conveyed instantly, even in situations where visibility or communication is limited. Furthermore, the investigation aims to enable wireless communication using the built-in capabilities of the microcontroller, allowing real-time data transmission to remote monitoring systems such as mobile devices or control rooms. This feature enhances coordination and enables supervisors to monitor the status of firefighters continuously.

Another objective of this investigation is to design a user-friendly interface using an OLED display to present real-time data, including environmental conditions, health parameters, and system status. This allows the firefighter to easily access important information during operations. The study also emphasizes efficient power management to ensure that the system can operate reliably for extended periods without frequent interruptions.

Overall, the objective of the present investigation is to develop a compact, reliable, and cost-effective system that integrates multiple functionalities into a single platform. By combining environmental sensing, health monitoring, motion detection, alert mechanisms, and communication, the proposed system aims to provide a comprehensive solution that significantly improves firefighter safety and effectiveness in indoor firefighting scenarios.

3.3 SCOPE OF THE PROJECT

The scope of the present project is centered on the design, development, and implementation of an IoT-based indoor firefighter navigation and safety suit that aims to enhance the safety and operational efficiency of firefighters working in hazardous environments. This project focuses on creating a compact and integrated wearable system that combines multiple sensing, processing, and communication technologies into a single platform. The system is intended to continuously monitor both environmental conditions and the physiological status of the firefighter, providing real-time data and alerts to ensure timely action during emergency situations.

The project primarily involves the integration of various sensors such as a gas sensor for detecting smoke and harmful gases, a temperature and humidity sensor for monitoring environmental conditions, a motion sensor for detecting movement and orientation, and a pulse oximeter sensor for measuring heart rate and blood oxygen levels. These sensors are interfaced with a microcontroller unit that processes the collected data and determines whether the conditions are within safe limits. Based on this analysis, the system generates alerts using visual and audio indicators such as LEDs and a buzzer to immediately notify the firefighter in case of danger.

In addition to monitoring and alerting, the project scope includes the implementation of location tracking using a GPS module, which provides approximate position information that can assist rescue teams in locating firefighters during emergencies. The system also incorporates wireless communication capabilities, enabling the transmission of real-time data to remote monitoring platforms such as mobile devices or control rooms. This feature enhances situational awareness and supports better coordination during firefighting operations.

Another important aspect within the scope of the project is the development of a user interface using an OLED display, which presents real-time information about environmental conditions, health parameters, and system status. This ensures that the firefighter has immediate access to critical data while operating in challenging conditions. The project also considers power management by utilizing a battery-based power supply system, ensuring that the device can function reliably during extended operations.

The scope of this project is limited to the development of a working prototype that demonstrates the feasibility and effectiveness of the proposed system. It does not include large-scale deployment, industrial-grade hardware design, or advanced technologies such as artificial intelligence-based decision-making or fully accurate indoor navigation systems. Additionally, while the system provides approximate location tracking, it does not implement highly precise indoor positioning techniques due to complexity and infrastructure requirements.

Overall, the project focuses on delivering a cost-effective, efficient, and reliable solution that integrates key safety features into a single wearable device. By addressing critical aspects such as hazard detection, health monitoring, motion analysis, alert generation, and communication, the system aims to significantly improve firefighter safety and support effective emergency response in indoor environments.

3.4 METHODOLOGY

The methodology adopted for this project focuses on the systematic design, integration, and implementation of an IoT-based indoor firefighter navigation and safety suit. The approach begins with identifying the key challenges faced by firefighters in hazardous indoor environments, such as exposure to toxic gases, high temperatures, limited visibility, and lack of real-time health monitoring. Based on these challenges, a comprehensive system is designed by integrating multiple sensors, a processing unit, communication modules, and alert mechanisms into a single wearable platform.

The first stage of the methodology involves the selection of appropriate hardware components that can effectively monitor both environmental and physiological conditions. Sensors such as the MQ-2 gas sensor are used to detect smoke and harmful gases, while the temperature and humidity sensor is used to measure environmental conditions. A pulse oximeter sensor is incorporated to monitor the firefighter's heart rate and blood oxygen levels, providing insight into their physical condition during operations. Additionally, the MPU6050 sensor is used to detect motion, orientation, and sudden movements, which helps in identifying situations such as falls or lack of activity. A GPS module is included to provide location tracking, enabling rescue teams to determine the approximate position of the firefighter.

In the next stage, all the selected sensors are interfaced with the ESP32 microcontroller, which serves as the central processing unit of the system. The ESP32 is chosen due to its built-in Wi-Fi and Bluetooth capabilities, which allow for efficient wireless communication. The microcontroller continuously collects data from all connected sensors and processes it in real time. Threshold values are defined for each parameter, and whenever the sensed values exceed safe limits, the system identifies it as a hazardous condition.

The methodology further includes the implementation of an alert mechanism to ensure immediate notification in case of danger. When abnormal conditions such as high gas concentration, extreme temperature, low oxygen levels, or abnormal heart rate are detected, the system activates a buzzer and LED indicators to alert the firefighter. This ensures that the user is immediately aware of any potential risk, even in situations where visibility is poor or external communication is limited.

Another important aspect of the methodology is the integration of a display unit, specifically an OLED display, which provides real-time information about environmental conditions, health parameters, and system status. This allows the firefighter to monitor critical data directly on the device. In addition to local display, the system utilizes the wireless capabilities of the ESP32 to transmit data to remote monitoring platforms. This enables supervisors or control rooms to continuously track the status of firefighters and make informed decisions during emergency situations.

The methodology also incorporates motion analysis using the MPU6050 sensor to detect abnormal patterns such as sudden falls or prolonged inactivity. When such events are detected, the system automatically generates an alert, which can be transmitted to remote monitoring systems. This feature plays a crucial role in ensuring timely assistance to firefighters who may be in distress.

Power management is another critical component of the methodology. The system is designed to operate using a battery power supply, supported by a buck converter to regulate voltage levels for different components. Efficient power utilization is considered to ensure continuous operation during firefighting missions without frequent interruptions.

Finally, the entire system is assembled into a compact and wearable prototype that can be integrated into a firefighter's safety suit. The prototype is tested under simulated conditions to evaluate its performance in detecting hazards, monitoring health parameters, and generating alerts. The results obtained from testing help in validating the effectiveness and reliability of the system.

Overall, the methodology follows a structured approach that includes component selection, system integration, real-time data processing, alert generation, communication, and testing. This ensures the development of a reliable and efficient IoT-based safety system that enhances the protection and operational capability of firefighters in indoor environments.

3.5 PROPOSED SYSTEM DESCRIPTION

The proposed system is an IoT-based indoor firefighter navigation and safety suit designed to enhance the safety, awareness, and operational efficiency of firefighters working in hazardous environments. The system integrates multiple sensors, a microcontroller, communication modules, and alert mechanisms into a compact wearable platform that continuously monitors both environmental conditions and the physiological status of the firefighter. By combining these features, the system provides real-time information and immediate alerts, enabling firefighters to respond quickly to dangerous situations and reducing the risk of injury or fatality.

At the core of the proposed system is the ESP32 microcontroller, which acts as the central processing unit. It is responsible for collecting data from all connected sensors, processing the information, and making decisions based on predefined threshold values. The ESP32 is selected due to its high processing capability and built-in Wi-Fi and Bluetooth features, which enable efficient wireless communication with remote monitoring systems. The system continuously gathers data from sensors such as the MQ-2 gas sensor, which detects smoke and harmful gases, and the temperature and humidity sensor, which monitors environmental conditions. These parameters are critical in identifying fire hazards and ensuring that firefighters are aware of the surrounding environment.

In addition to environmental monitoring, the system incorporates a pulse oximeter sensor to track the firefighter's physiological condition, including heart rate and blood oxygen levels. This feature is essential for identifying signs of physical stress, fatigue, or potential medical emergencies during operations. The MPU6050 sensor is used to monitor motion and orientation, enabling the system to detect abnormal conditions such as falls or lack of movement. When such events are detected, the system can automatically generate alerts, ensuring that immediate assistance can be provided.

The proposed system also includes a GPS module for location tracking, which helps in determining the approximate position of the firefighter. Although GPS has limitations in indoor environments, it can still provide useful information when combined with other sensor data. This feature is particularly important for rescue operations, as it allows teams to locate firefighters who may be trapped or in distress.

To ensure immediate notification of dangerous conditions, the system is equipped with alert mechanisms such as a buzzer and LED indicators. When the sensed parameters exceed safe limits, the system activates these alerts to warn the firefighter. This is especially useful in low-visibility environments where quick awareness is crucial. Additionally, an OLED display is integrated into the system to provide real-time information about environmental conditions, health parameters, and system status, allowing the firefighter to monitor key data directly.

Wireless communication plays a vital role in the proposed system, as it enables the transmission of real-time data to remote monitoring platforms such as control rooms or mobile devices. This allows supervisors to track the status of firefighters, monitor environmental conditions, and make informed decisions during emergency situations. The ability to access real-time data enhances coordination and improves the overall effectiveness of firefighting operations.

The system is designed to be powered by a battery, with a buck converter used to regulate voltage levels and ensure stable operation of all components. Care is taken to optimize power consumption so that the system can operate reliably for extended periods. The entire setup is integrated into a wearable form, making it convenient for firefighters to use without hindering their movement or performance.

Overall, the proposed system provides a comprehensive solution for firefighter safety by integrating hazard detection, health monitoring, motion analysis, location tracking, alert generation, and wireless communication into a single platform. By addressing the limitations of existing systems, this solution enhances situational awareness, reduces response time, and significantly improves the safety and survivability of firefighters in indoor fire emergencies.

3.5.1 Block Diagram

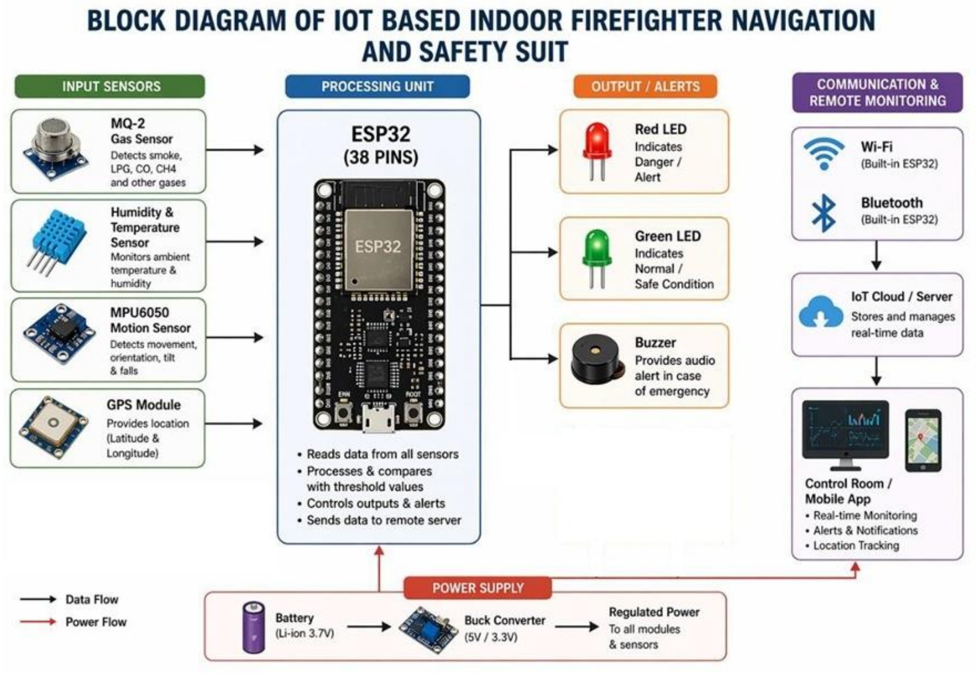

Fig.3.1: FIREFIGHTER NAVIGATION & SAFETY SUIT Block Diagram

This Fig.3.1 represents an **IoT-based Indoor Firefighter Navigation and Safety Suit** that helps protect firefighters during rescue operations. The system uses various input sensors such as the **MQ-2 gas sensor** to detect smoke and harmful gases, a **temperature and humidity sensor** to monitor surroundings, an **MPU6050 motion sensor** to detect movement or falls, a **pulse oximeter** to measure heart rate and oxygen level, and a **GPS module** for tracking location. These sensors continuously collect important environmental and health data.

All sensor data is sent to the **ESP32 microcontroller**, which processes the information and compares it with preset safety limits. If danger is detected, the system activates output devices like a **red LED**, **green LED**, **buzzer**, and **OLED display** to provide warnings and live status updates. Using **Wi-Fi and Bluetooth**, the data is also transmitted to the **IoT cloud server**, control room, or mobile app for real-time monitoring, alerts, and location tracking. The system is powered by a battery with regulated power supply for all modules.

3.7 ADVANTAGES OF THE PROPOSED SYSTEM

The proposed IoT-based indoor firefighter navigation and safety suit offers several **and** significant advantages over traditional firefighting systems by integrating advanced sensing, monitoring, and communication technologies into a single wearable platform. One of the major advantages of this system is the ability to provide real-time monitoring of both environmental and physiological conditions. By continuously tracking parameters such as gas concentration, temperature, humidity, heart rate, and oxygen levels, the system ensures that firefighters are always aware of their surroundings as well as their physical condition. This real-time awareness helps in identifying dangerous situations at an early stage, thereby reducing the risk of severe injuries or fatalities.

Another important advantage of the proposed system is its integrated design, which combines multiple functionalities into a single compact unit. Unlike existing systems that operate independently, this system brings together hazard detection, health monitoring, motion analysis, location tracking, and communication in one platform. This integration simplifies operation, reduces the need for multiple devices, and enhances overall efficiency during firefighting operations. The centralized processing using the ESP32 microcontroller ensures that all sensor data is analyzed in real time and appropriate actions are taken immediately.risk of severe injuries or fatalities.

The system also provides effective and instant alert mechanisms, which play a crucial role in ensuring firefighter safety. The use of a buzzer for audio alerts and LEDs for visual indication ensures that warnings are delivered promptly, even in low-visibility and high-noise environments. This immediate notification allows firefighters to respond quickly to dangerous conditions, such as the presence of toxic gases, high temperatures, or abnormal health parameters.

Another key advantage is the inclusion of motion detection using the MPU6050 sensor, which enables the system to identify situations such as falls or prolonged inactivity. This feature is particularly important in detecting emergencies where the firefighter may be unconscious or unable to seek help. Automatic alerts generated in such scenarios ensure that rescue teams can take immediate action, thereby improving response time and increasing the chances of survival.

The proposed system also enhances communication and coordination through wireless data transmission. By utilizing the built-in Wi-Fi and Bluetooth capabilities of the ESP32, the system can send real-time data to remote monitoring platforms such as control rooms or mobile devices. This allows supervisors to continuously monitor the status of firefighters and make informed decisions during emergency situations. The inclusion of GPS-based location tracking further supports rescue operations by providing approximate location details of the firefighter .

Another advantage of the system is the use of an OLED display, which provides a clear and real-time visualization of important parameters. This enables firefighters to easily access critical information without relying solely on external monitoring systems. The display enhances situational awareness and helps in making quick decisions during operations .

Power efficiency and portability are also notable advantages of the proposed system. The use of a battery-powered setup with proper voltage regulation ensures reliable operation for extended periods. The compact and wearable design of the system makes it easy to integrate into a firefighter's suit without restricting movement or causing discomfort.

Additionally, the proposed system is cost-effective and scalable, as it utilizes commonly available componzents and technologies. This makes it suitable for implementation in various environments, including resource-constrained settings. The modular design of the system also allows for future enhancements, such as the integration of advanced sensors or intelligent algorithms .

Overall, the proposed system offers a comprehensive and efficient solution for improving firefighter safety by providing real-time monitoring, quick alert mechanisms, reliable communication, and enhanced situational awareness. These advantages make it a highly effective system for use in indoor firefighting operations, where safety and timely response are of utmost importance.

CHAPTER 4
EXPERIMENTAL METHODS

4.1 EXPERIMENTAL OVERVIEW

The experimental work for this project focuses on the design, implementation, and evaluation of the IoT-based indoor firefighter navigation and safety suit under controlled conditions. The objective of the experimental phase is to validate the functionality, reliability, and effectiveness of the proposed system in detecting hazardous conditions, monitoring physiological parameters, and generating timely alerts. The experimental setup is developed as a prototype model that integrates all the selected components, including sensors, microcontroller, communication modules, and output devices, into a single wearable system.

The experimental process begins with the assembly of hardware components, where sensors such as the MQ-2 gas sensor, temperature and humidity sensor, MPU6050 motion sensor, pulse oximeter sensor, and GPS module are interfaced with the ESP32 microcontroller. Each component is connected according to the required pin configuration, and proper power supply is ensured using a regulated battery source. The OLED display, LEDs, and buzzer are also connected to provide real-time output and alerts. Once the hardware setup is completed, the system is programmed using embedded C/C++ in the Arduino IDE, where sensor data acquisition, processing, and decision-making algorithms are implemented.

The experimental evaluation involves testing the system under various simulated conditions that represent real firefighting scenarios. Environmental testing is carried out by exposing the gas sensor to smoke or gas sources to observe its response and detection capability. Temperature and humidity variations are introduced to evaluate the accuracy and responsiveness of the respective sensor. Physiological monitoring is tested by measuring heart rate and oxygen levels using the pulse oximeter sensor to ensure correct data acquisition and display. Motion detection is validated by simulating movements, falls, and periods of inactivity to verify whether the MPU6050 sensor accurately detects these conditions and triggers alerts.

In addition to sensor-level testing, the system's alert mechanisms are evaluated to ensure that the buzzer and LED indicators respond appropriately when abnormal conditions are detected. The OLED display is tested for its ability to present real-

time data clearly and accurately. Wireless communication is also assessed by transmitting data from the ESP32 to a remote device or monitoring platform, verifying the reliability and consistency of data transfer. The GPS module is tested to confirm its ability to provide location coordinates, which can be used for tracking purposes.

The experimental methodology also includes testing the system’s overall performance by integrating all components and observing its behavior in combined scenarios. For example, simultaneous detection of high gas levels and abnormal heart rate is tested to evaluate how the system prioritizes and responds to multiple alerts. The power consumption of the system is monitored to ensure efficient energy usage and stable operation over extended periods.

The results obtained from these experiments are analyzed to assess the accuracy, responsiveness, and reliability of the proposed system. Any discrepancies or limitations observed during testing are noted and used for further improvement. The experimental overview demonstrates that the system is capable of providing real-time monitoring, detecting hazardous conditions, and generating alerts effectively, thereby validating its suitability for enhancing firefighter safety in indoor environments.

4.2HARDWARE CIRCUIT DIAGRAM

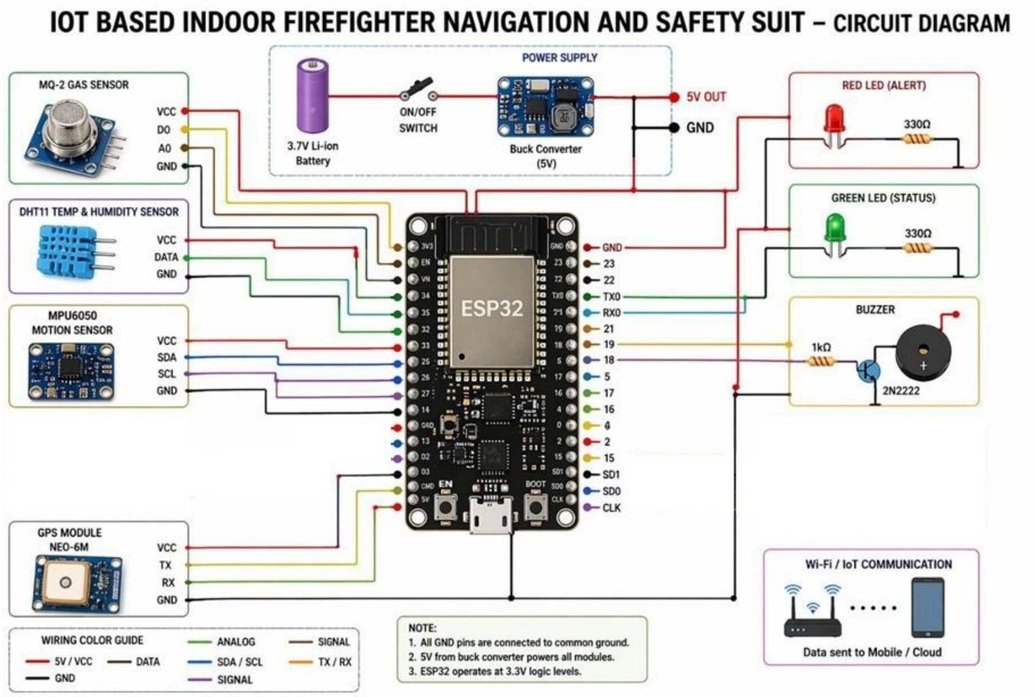

Fig:4.2. Firefighter Navigation & Safety Suit Circuit Diagram

This Fig.4.2 shows an IoT-based Indoor Firefighter Navigation and Safety Suit built around an ESP32 microcontroller, which acts as the central processing unit connecting all sensors and output devices. The system is powered by a 3.7V Li-ion battery, which is stepped up/stabilized to 5V using a buck converter, and then distributed to different modules while the ESP32 operates at 3.3V logic levels. On the sensing side, it integrates a DHT11 sensor for monitoring temperature and humidity, an MQ-2 gas sensor for detecting smoke and harmful gases, and an MPU6050 motion sensor for tracking movement, tilt, and fall detection inside hazardous environments. A pulse oximeter (MAX30100) is included to monitor the firefighter's heart rate and oxygen levels, ensuring physiological safety, while a GPS module (NEO-6M) provides real-time location tracking for navigation and rescue assistance. For output and alerts, the system uses a red LED for danger alerts, a green LED for normal status, and a buzzer circuit driven through a transistor (2N2222) for strong audible warning signals. A 0.96-inch OLED display (I2C interface) shows real-time sensor readings, status updates, and navigation data, while Wi-Fi connectivity enables IOT-based data transmission to mobile or cloud platforms for remote monitoring.

4.3FLOW CHART

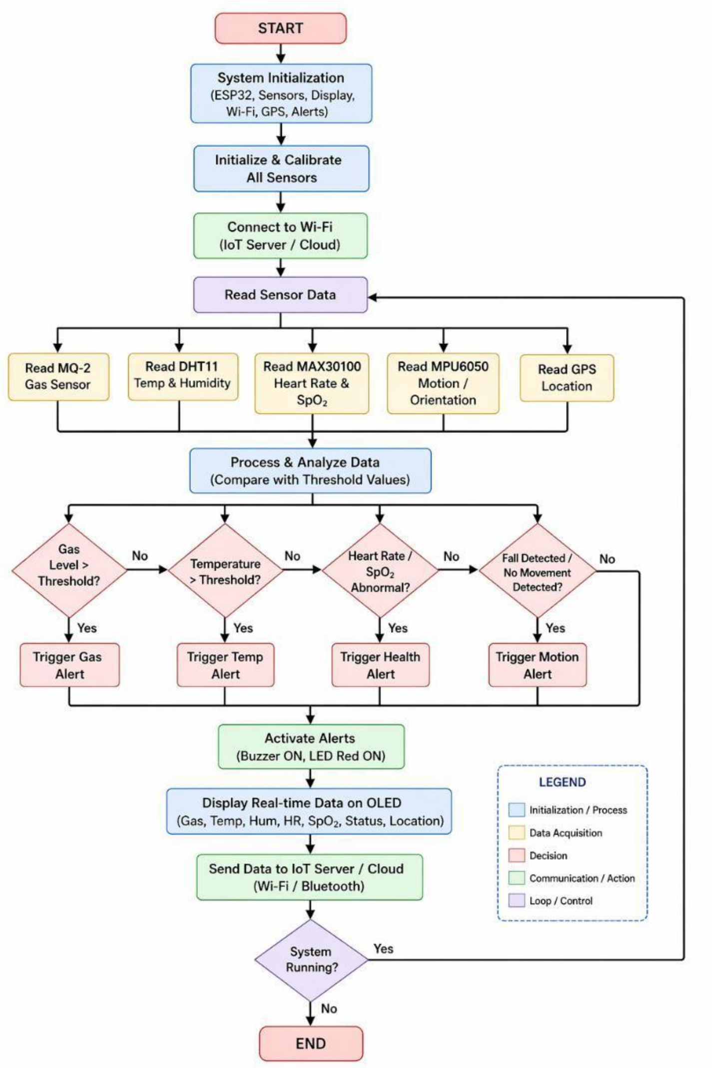

Fig:4.3: Flowchart of Overall System Operation

The flowchart illustrates the working sequence of the proposed IoT-based indoor firefighter navigation and safety system, beginning with the initialization of all sensors and system components. Once the system is powered on, the ESP32 microcontroller continuously acquires data from various sensors, including the MQ-2 gas sensor, temperature and humidity sensor, MPU6050 motion sensor, pulse oximeter sensor, and GPS module. The collected data is then processed in real time to evaluate environmental conditions and the physiological status of the firefighter. Based on predefined threshold values, the system checks whether the sensed parameters fall within safe limits. If all conditions are normal, the system continues monitoring and displays real-time data on the OLED display.

4.4 EXPERIMENTAL SETUP

The experimental setup of the proposed IoT-based indoor firefighter navigation and safety system is designed to evaluate the functionality and performance of all integrated components under controlled conditions. The system consists of hardware components such as the ESP32 microcontroller, MQ-2 gas sensor, temperature and humidity sensor, MPU6050 motion sensor, pulse oximeter sensor, GPS module, OLED display, LEDs, buzzer, and a regulated power supply. All components are interconnected using appropriate wiring based on their pin configurations, ensuring stable communication and proper operation.

The ESP32 microcontroller acts as the central unit and is programmed using the Arduino IDE. It is responsible for reading sensor values, processing data, and controlling output devices. The MQ-2 sensor is placed in proximity to gas sources during testing to detect smoke and harmful gases. The temperature and humidity sensor is positioned to measure ambient environmental conditions. The pulse oximeter sensor is used to monitor physiological parameters, while the MPU6050 sensor is fixed in a way that allows accurate detection of motion and orientation. The GPS module is connected to receive satellite signals for location tracking, and the OLED display is interfaced to present real-time data. The buzzer and LEDs are connected as output devices to provide alerts during abnormal conditions.

The system is powered using a battery source, and a voltage regulator or buck converter is used to ensure that all components receive appropriate voltage levels. The entire setup is assembled in a compact form to simulate a wearable device, making it suitable for practical firefighter applications.

4.5 EXPERIMENTAL PROCEDURE

The experimental procedure is carried out in a systematic manner to evaluate the performance of individual components as well as the overall system. Initially, the system is powered on, and all sensors are initialized. The ESP32 begins continuous data acquisition from all sensors, and the values are displayed on the OLED screen for monitoring.

For gas detection testing, the MQ-2 sensor is exposed to smoke or gas sources, and the system's response is observed. The sensor readings are monitored to ensure that the system accurately detects hazardous gas levels and triggers alerts when the threshold is exceeded. Similarly, temperature and humidity testing is performed by exposing the sensor to varying environmental conditions to evaluate its accuracy and responsiveness.

The pulse oximeter sensor is tested by placing a finger on the sensor to measure heart rate and oxygen saturation levels. The readings are observed for consistency and correctness. Motion detection is tested by simulating different scenarios such as normal movement, sudden falls, and periods of inactivity. The MPU6050 sensor data is analyzed to verify whether the system correctly identifies abnormal conditions and generates alerts.

The GPS module is tested by placing the system in an open environment to receive satellite signals. The latitude and longitude values are recorded and verified for accuracy. Wireless communication is tested by transmitting sensor data from the ESP32 to a remote monitoring platform, ensuring reliable data transfer.

Finally, the entire system is tested under combined conditions where multiple parameters change simultaneously. This helps in evaluating the system's ability to handle real-time scenarios and generate appropriate alerts.

4.5.1 ESP32 (38-PIN MICROCONTROLLER)

The proposed IoT-based firefighter safety system uses an ESP32 (38-pin) as the main controller to collect, process, and transmit data from all sensors. The MQ-2 sensor detects smoke and harmful gases, while the temperature and humidity sensor monitors environmental conditions. The MPU6050 detects motion, falls, and inactivity, and the pulse oximeter sensor measures heart rate and oxygen levels to track the firefighter's health. A GPS module provides location data for tracking and rescue. The OLED display shows real-time information, while a buzzer and red/green LEDs give alerts for unsafe and safe conditions. The system is powered by a battery, making it compact and suitable for wearable use.

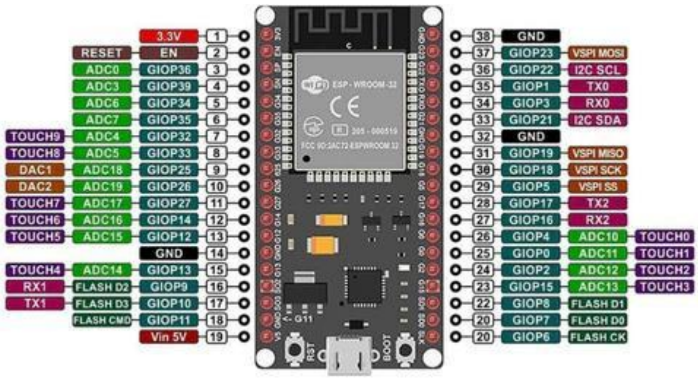

Fig:4.5.1: ESP32 (38-Pin Microcontroller)

4.5.2 ACCELEROMETER SENSOR(MPU6050)

The accelerometer sensor (MPU6050) is used in this project to detect the movement and orientation of the firefighter by sensing acceleration and angular changes. It helps in identifying motion patterns such as normal movement, sudden falls, or lack of activity, which may indicate that the firefighter is in danger or unconscious. This sensor plays an important role in ensuring safety by providing real-time motion data and enabling the system to generate alerts during abnormal conditions. The MPU6050 is reliable, compact, and enhances the overall effectiveness of the system when combined with other sensors.

Fig:4.5.3: Accelerometer Sensor(MPU6050)

4.5.3 MQ-2 GAS SENSOR

The MQ-2 gas sensor is used in this project to detect the presence of smoke and harmful gases such as LPG, methane, and carbon monoxide in the environment. It works by sensing changes in gas concentration and converting them into electrical signals that can be processed by the microcontroller. This helps in identifying dangerous conditions during fire incidents, allowing the system to alert the firefighter immediately. The sensor plays a crucial role in ensuring safety by providing early detection of hazardous gases. It is chosen because it is sensitive, reliable, cost-effective, and suitable for continuous monitoring.

Fig:4.5.4: MQ-2 Gas Sensor

4.5.4 GPS MODULE

The GPS module is used in this project to track the location of the firefighter by providing real-time latitude and longitude coordinates. It communicates with satellites to determine the position, which can be used by rescue teams to locate firefighters during emergency situations. This feature is especially useful when a firefighter is trapped or in distress, as it helps in quick response and rescue operations. The GPS module is reliable, easy to interface with the microcontroller, and enhances the overall safety system by enabling location tracking.

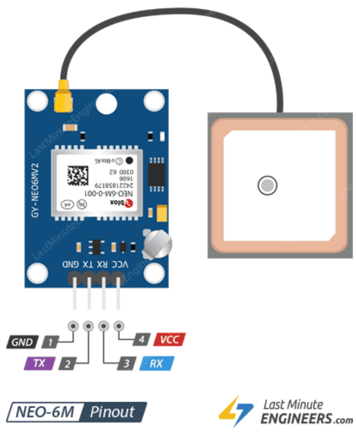

Fig:4.5.5: GPS Module

4.5.5 HUMIDITY SENSOR(DHT11)

The DHT11 sensor is used in this project to measure the temperature and humidity of the surrounding environment. It works by sensing changes in atmospheric conditions and converting them into digital signals that canmicrocontroller. This helps in identifying high-temperature conditions and changes in humidity levels during fire incidents, which are important indicators of environmental danger. The sensor plays a key role in monitoring the surroundings and ensuring firefighter safety. It is chosen because it is simple to use, low-cost, and suitable for continuous environmental monitoring.

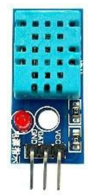

Fig:4.5.6: HUMIDITY Sensor(DHT11)

4.5.6 BUZZER

The 2-pin buzzer is used in this project as an alert device to provide audible warnings during dangerous conditions. It is activated by the microcontroller when abnormal parameters such as high gas levels, extreme temperature, or health issues are detected. The buzzer produces a loud sound, ensuring that the firefighter is immediately alerted even in low-visibility or noisy environments. This helps in quick response and increases safety during emergency situations.

Fig:4.5.7: BuzzerRED LED

The red LED is used in this project as a visual indicator to signal dangerous or abnormal conditions. It is activated by the microcontroller when parameters such as high gas levels, extreme temperature, or unsafe health readings are detected. The glowing red light provides a clear warning to the firefighter, especially in situations where audio alerts may not be sufficient. This helps in quickly identifying hazardous conditions and taking immediate action. The red LED is simple, low-power, and effective for visual alert indication.

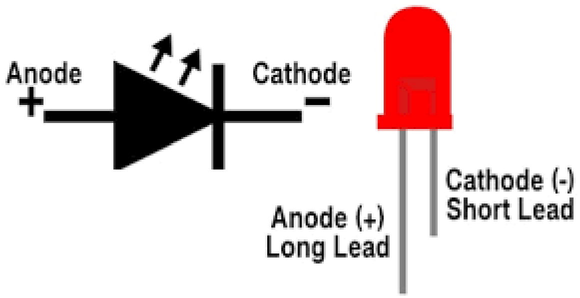

Fig:4.6: Red LED

4.5.7 GREEN LED

The green LED is used in this project as a visual indicator to show normal and safe operating conditions. It is activated by the microcontroller when all monitored parameters such as gas levels, temperature, and health readings are within safe limits. The glowing green light provides reassurance to the firefighter that the system is functioning properly and no immediate danger is detected. It is simple, energy-efficient, and useful for continuous status indication.

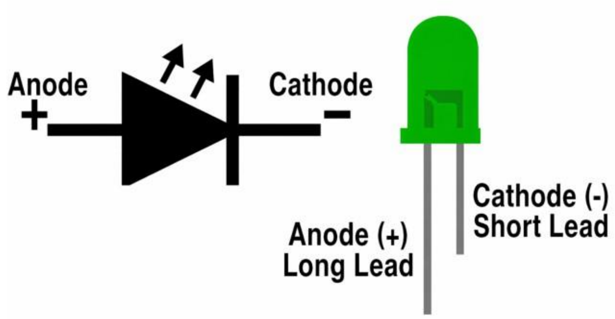

Fig:4.6.1: Green LED

4.6

4.7 SOFTWARE DEVELOPMENT TOOLS

The software development for this project is carried out using the Arduino IDE, which provides a simple and efficient platform for writing, compiling, and uploading code to the ESP32 microcontroller. The programming is done using Embedded C/C++, enabling easy integration of various sensors and modules. Required libraries for components such as the MPU6050, OLED display, pulse oximeter sensor, and GPS module are included to simplify communication and data processing. The Arduino IDE also offers a serial monitor for debugging and testing, allowing real-time observation of sensor values and system behavior. This development environment is chosen because it is user-friendly, widely supported, and suitable for rapid prototyping of IoT-based embedded systems.

4.7.1 ARDUINO IDE

The Arduino IDE is used in this project as the software platform to write, compile, and upload the program code to the ESP32 microcontroller. It provides a simple and user-friendly interface where the logic for sensor data collection, processing, alert generation, and communication is implemented using embedded C/C++ language. The IDE supports various libraries required for interfacing components such as the OLED display, sensors, and communication modules. It also allows real-time debugging and serial monitoring, which helps in testing and troubleshooting the system. The Arduino IDE is chosen because it is easy to use, widely supported, and suitable for developing IoT-based embedded systems.

4.7.2 BLYNK

The Blynk platform is used in this project to enable real-time remote monitoring and control of the firefighter safety system through a mobile application.

It allows sensor data such as gas levels, temperature, heart rate, and location to be displayed on a smartphone, providing instant access to critical information from anywhere. The ESP32 communicates with the Blynk app using Wi-Fi, enabling continuous data transmission and alert notifications during abnormal conditions. This helps supervisors or rescue teams monitor the firefighter’s status in real time and take quick action when needed. Blynk is chosen because it is easy to use, supports IoT integration, and provides a reliable interface for remote monitoring.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 INTRODUCTION

This chapter presents the results obtained from the experimental evaluation of the proposed IoT-based indoor firefighter navigation and safety system and provides a detailed discussion of its performance. The primary objective of this chapter is to analyze how effectively the system monitors environmental and physiological parameters, detects hazardous conditions, and generates timely alerts. The system was tested under various simulated conditions to evaluate its accuracy, responsiveness, and reliability in real-time operation.

The results include observations from individual sensor performance, integrated system behavior, alert mechanisms, and wireless communication. Parameters such as gas detection, temperature and humidity monitoring, heart rate and oxygen level measurement, motion detection, and location tracking were carefully analyzed. The system's ability to respond to abnormal conditions, such as high gas concentration, extreme temperature, sudden falls, and health irregularities, is also evaluated in this chapter.

In addition to presenting the results, this chapter discusses the effectiveness of the system in improving firefighter safety. The advantages of integrating multiple sensors into a single wearable platform are examined, along with the system's ability to provide real-time monitoring and alerts. Any limitations observed during testing, such as sensor sensitivity variations or GPS performance in indoor environments, are also discussed.

Overall, this chapter provides a comprehensive evaluation of the proposed system, demonstrating its capability to enhance situational awareness, reduce response time, and improve safety for firefighters operating in hazardous indoor conditions.

5.2 RESULTS

The proposed IoT-based indoor firefighter navigation and safety system was successfully implemented and tested under various simulated conditions to evaluate its performance. The system demonstrated effective real-time monitoring of environmental and physiological parameters. The MQ-2 gas sensor responded accurately to the presence of smoke and harmful gases, triggering alerts when the gas concentration exceeded predefined threshold levels. The DHT11 sensor provided consistent readings of temperature and humidity, allowing the system to identify high-temperature conditions associated with fire hazards.

The pulse oximeter sensor successfully measured heart rate and blood oxygen (SpO_2) levels, providing reliable data for monitoring the firefighter's health status. The MPU6050 sensor accurately detected motion, including normal movement, sudden falls, and inactivity, enabling the system to identify emergency situations. The GPS module provided location coordinates, which were useful for tracking purposes, although minor variations were observed in indoor conditions.

The OLED display effectively showed real-time data, ensuring that all parameters were visible to the user. The buzzer and LED indicators responded immediately when abnormal conditions were detected, confirming the effectiveness of the alert system. Additionally, the ESP32 successfully transmitted data to the Blynk platform via Wi-Fi, enabling remote monitoring of all parameters in real time.

Overall, the system performed as expected, demonstrating its capability to detect hazards, monitor health conditions, and provide timely alerts.

5.3 DISCUSSION

The results obtained from the experimental analysis indicate that the proposed system is effective in enhancing firefighter safety through real-time monitoring and alert mechanisms. The integration of multiple sensors into a single system provides comprehensive situational awareness, which is essential in hazardous environments. The MQ-2 sensor plays a crucial role in detecting harmful gases at an early stage, while the DHT11 sensor helps in identifying extreme environmental conditions. The pulse oximeter sensor adds an important dimension by monitoring the firefighter's health, which is often overlooked in traditional systems.

The motion detection feature using the MPU6050 sensor significantly improves safety by identifying falls or inactivity, which are critical emergency scenarios. The alert system, consisting of a buzzer and LEDs, ensures immediate notification, allowing the firefighter to take quick action. The OLED display enhances usability by providing clear and real-time information directly on the device.

Wireless communication using the ESP32 and Blynk platform further improves the effectiveness of the system by enabling remote monitoring. This allows supervisors or rescue teams to track the firefighter's condition and location, leading to faster response times. However, some limitations were observed during testing. The GPS module showed reduced accuracy in indoor environments, which is a common limitation due to signal obstruction. Additionally, sensor readings may slightly vary depending on environmental conditions.

Despite these minor limitations, the system demonstrates strong performance in terms of accuracy, responsiveness, and reliability. The integration of multiple features into a single wearable device makes the system highly efficient and practical for real-world applications. Overall, the proposed system provides a significant improvement over existing methods by offering a comprehensive and intelligent solution for firefighter safety.

5.4 RESULT TABLES (WITH VALUES)

Table 5.4.1: Environmental Monitoring Results

Time (s)	Temperature (°C)	Humidity (%)	Gas Level (ppm)	Status
0	30	55	120	Safe
10	35	58	180	Safe
20	42	60	300	Alert
30	50	65	450	Danger
40	55	70	600	Danger

Table 5.4.2: Motion Detection Results

Time (s)	Movement Status	System Response
0	Normal	No Alert
10	Walking	No Alert
20	Sudden Tilt	Warning
30	Fall Detected	Alert Triggered
40	No Movement	Emergency Alert

5.5 HARDWARE IMPLEMENTATION

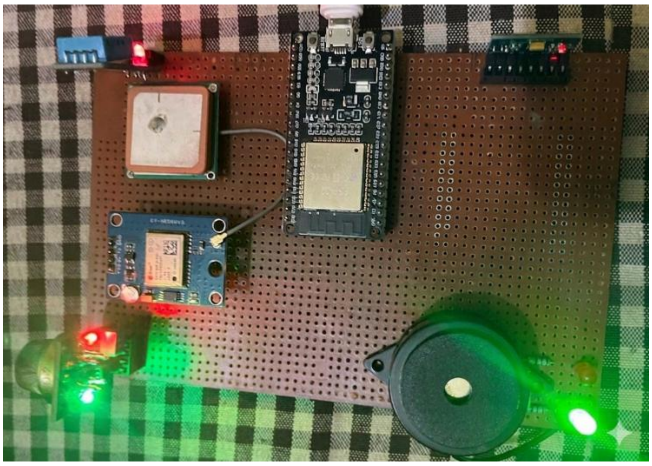

Fig:5.5HARDWARE IMPLEMENTATION

This diagram shows a prototype of a firefighter safety monitoring system assembled on a perforated board, where a central microcontroller (likely an ESP32) controls and coordinates multiple sensors and output devices. The setup includes a GPS module for real-time location tracking, along with environmental sensors that measure parameters such as temperature and gas levels. An OLED display presents this information clearly, showing values like temperature (around 30.4°C), gas concentration, and motion data from an accelerometer or gyroscope, along with a status message indicating whether conditions are safe. Visual indicators such as red and green LEDs provide quick status feedback, while a buzzer or speaker is included to trigger audible alerts if dangerous conditions are detected. Overall, the system continuously gathers and processes environmental and motion data to help ensure the safety of a firefighter by providing real-time monitoring, alerts, and location tracking.

CHAPTER 6

SUMMARY AND CONCLUSION

6.1 SUMMARY

Fire accidents and hazardous environments pose extreme risks to firefighters, often making rescue operations highly dangerous and unpredictable. Traditional firefighting methods rely heavily on manual observation and communication, which may not always provide timely and accurate information. To overcome these limitations, this project presents an IoT-based Fire Fighter Navigation System using ESP32-WROOM-32, designed to improve safety, monitoring, and decision-making during emergency situations.

The proposed system integrates multiple sensors and communication technologies into a single wearable or portable unit for firefighters. The core controller, ESP32, plays a central role in collecting, processing, and transmitting real-time data from various sensors. The system is designed to continuously monitor environmental conditions as well as the physiological status of the firefighter.

The DHT11 sensor is used to measure ambient temperature and humidity, which helps in identifying fire intensity and surrounding heat conditions. The MQ-2 gas sensor detects harmful gases and smoke levels, providing early warnings of toxic environments. The MPU6050 sensor is used for motion and tilt detection, which helps in identifying falls, sudden movements, or disorientation of firefighters inside buildings.

In addition, the GPS module provides real-time location tracking, which is essential for guiding rescue teams and monitoring firefighter movement in disaster zones. The heart rate sensor is used to monitor the physiological condition of the firefighter, ensuring that fatigue or stress levels can be detected early. All sensor data is continuously processed by the ESP32 microcontroller and displayed locally on a 0.96-inch OLED display, allowing firefighters to view critical information instantly.

For alert mechanisms, the system uses a combination of a buzzer, LED indicators (red and green), and a vibration motor. These components provide or abnormal body conditions are detected. This ensures that firefighters can take quick action even in low-visibility or high-stress environments.

One of the major features of this system is its IoT-based remote monitoring capability. The ESP32 connects to a Wi-Fi network and sends real-time sensor data to the Blynk IoT platform. This allows control room operators or team leaders to monitor the condition and location of firefighters remotely using a smartphone or computer. This feature significantly improves coordination and response time during emergencies.

The system also supports wireless communication using HC-05 Bluetooth module, which can be used for short-range data transfer or backup communication. The integration of multiple technologies ensures that the system remains reliable even if one communication method fails.

From a design perspective, the system is built to be compact, energy-efficient, and wearable, making it suitable for real-life firefighting operations. The use of a buck converter and battery power supply ensures stable voltage and portability, allowing the system to operate in remote and harsh environments.

The developed system significantly improves firefighter safety by reducing dependency on manual communication and observation. Real-time data monitoring ensures that hazardous conditions are detected early, allowing timely evacuation or support. The combination of environmental sensing, health monitoring, and GPS tracking makes the system a comprehensive solution for modern firefighting challenges.

This project demonstrates how IoT and embedded systems can be effectively used in real-world emergency applications. It bridges the gap between firefighters and control centers by enabling continuous data exchange and intelligent alerting.

In conclusion, the Fire Fighter Navigation System using IoT is a reliable and efficient safety solution designed to assist firefighters in dangerous environments. The system successfully integrates multiple sensors, wireless communication, and real-time monitoring into a single platform. It enhances situational awareness, improves decision-making, and ensures better coordination during rescue operations.

6.2 CONCLUSION

The Fire Fighter Navigation System using IoT demonstrates an effective and practical approach to improving firefighter safety in hazardous and unpredictable environments. The system successfully integrates the ESP32-WROOM-32 microcontroller with multiple sensors such as DHT11, MQ-2, MPU6050, GPS module, and heart rate sensor to continuously monitor environmental conditions and the physical status of firefighters in real time.

Through real-time data acquisition and processing, the system is capable of detecting critical situations such as high temperature, smoke or gas leakage, abnormal body movement, and distress conditions. The integration of alert mechanisms like buzzer, LED indicators, and vibration motor ensures immediate notification to the firefighter, enabling quick decision-making during emergencies.

The inclusion of IoT-based communication using the Blynk platform enhances the system by allowing remote monitoring of live sensor data and GPS location. This enables rescue teams and control centers to track firefighters efficiently and respond faster during critical situations. The system also improves coordination, reduces response time, and increases the chances of successful rescue operations.

Overall, this project proves to be a cost-effective, reliable, and scalable safety solution for modern firefighting applications. It effectively combines embedded systems and IoT technology to enhance situational awareness and operational safety. With further improvements such as AI-based prediction, long-range communication, and advanced sensors, the system can be developed into a more powerful real-world emergency management tool.

6.3 SOFTWARE IMPLEMENTATION

This diagram shows a monitoring dashboard for a firefighter that displays real-time environmental and physiological data collected from a connected device. At the top, the interface identifies the device labeled “Fire Fighter” along with its connection status, followed by quick control options and a time range selector that allows users to view data trends over different periods. The main section presents gauge-style indicators that visualize key parameters, including a gas level reading of 448 ppm indicating the concentration of gas in the environment, temperature

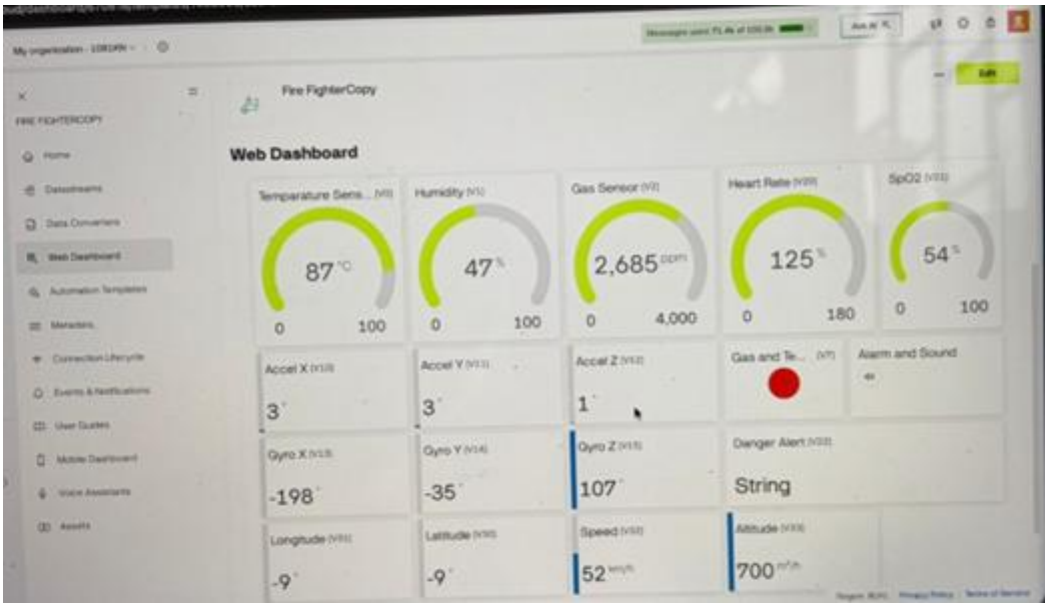

Fig:6.3 Software Implementation

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31.2°C representing the surrounding conditions, a humidity level of 78.5% showing the amount of moisture in the air, and an SpO2 value of 70% reflecting the firefighter’s blood oxygen level. Beneath these gauges, alert panels are displayed where the Temperature Alert and Gas Alert sections indicate whether predefined safety thresholds have been exceeded, currently showing zero, which means no alerts are active. Additionally, there are alarm and sound controls that can be used to trigger warnings if hazardous conditions arise. At the bottom of the dashboard, further sensor data such as acceleration along the X, Y, and Z axes and location details including longitude and latitude are shown.

6.4 FUTURE ENHANCEMENTS

The Fire Fighter Navigation System using IoT has strong potential for further enhancement and real-world deployment. Although the current system effectively integrates multiple sensors, ESP32-based processing, and IoT monitoring, several improvements can be made to increase its performance, reliability, and scalability in complex emergency environments.

One of the major future improvements is the integration of Artificial Intelligence (AI) and Machine Learning algorithms. By analyzing historical sensor data such as temperature, gas levels, and heart rate patterns, the system can predict fire intensity and detect dangerous conditions more accurately. This predictive capability can help in early warning and better decision-making during rescue operations.

The system can also be enhanced by incorporating thermal imaging cameras or infrared sensors, which will help firefighters see through smoke-filled environments. This will significantly improve visibility and navigation inside burning buildings where human vision is limited.

Another important improvement is the use of long-range communication technologies such as LoRa (Long Range) or Zigbee networks. These technologies can ensure reliable communication even in large-scale disaster areas where Wi-Fi or Bluetooth connectivity is weak or unavailable.

In future versions, the system can be connected to a cloud-based emergency response platform, where all firefighter data is stored and analyzed in real time. This will allow multiple control centers, hospitals, and disaster management teams to access critical information instantly, improving coordination and response speed.

Battery optimization is another key area for improvement. By using low-power hardware designs and energy-efficient algorithms, the system can achieve longer operational time, which is crucial during extended rescue missions.

The system can also be upgraded with voice communication modules or helmet-mounted smart displays, allowing firefighters to receive instructions directly without checking external devices. This will improve usability and safety during high-risk operations.

Additionally, integration with drone technology can be explored in the future. Drones equipped with cameras and sensors can assist in mapping fire zones, identifying trapped individuals, and supporting navigation for firefighters on the ground.

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ANNEXURES ANNEXURE 1: PO PSO MAPPING

Project Title: IoT-Based Indoor Firefighter Navigation and Safety Suit

PO PSO MAPPING TABLE:

PO-PSO	Mapping Level	Justification
PO1	3	Applied engineering knowledge of ESP32, sensors (DHT11, MQ-2, GPS, MPU6050) and embedded systems.
PO2	3	Identified and analyzed real-time fire safety problems and designed a monitoring solution.
PO3	3	Designed an integrated IoT-based firefighter safety system with alerts and tracking.
PO4	2	Conducted data analysis from sensors like heart rate, gas, and motion sensors.
PO5	3	Used Arduino IDE, Blynk IoT platform, and sensor interfacing tools effectively.
PO6	3	Improved firefighter safety and emergency response, contributing to societal safety.
PO7	2	Ensured ethical use of data for safety monitoring and emergency response.
PO8	3	System designed for team-based emergency rescue operations and coordination.

PO9	2	Used Blynk dashboard and reporting for communication of sensor data.
PO10	2	Managed system design, hardware integration, and software development efficiently.
PO11	3	Learned IoT, ESP32 programming, and real-time embedded system applications.
PSO1	1	Not directly related to power systems.
PSO2	1	Not applicable to electrical drives.
PSO3	3	Strongly applied embedded systems, sensors, and IoT-based real-time application development.

ANNEXURE 2: SUSTAINABLE DEVELOPMENT GOALS MAPPINGS

Project Title: IoT-Based Indoor Firefighter Navigation and Safety Suit

SDG MAPPING TABLE:

SDG Goal	Description	Mapped Program Outcome
SDG 8	Decent Work and Economic Growth	PO10
SDG 9	Industry, Innovation and Infrastructure	PO3, PO5
SDG 11	Sustainable Cities and Communities	PO6, PO3